

The Life of a Cash Bubble: Narrative & Visualization Framework

Companion piece to *The Cash Bubble Hypothesis: A Physical Theory of Personal Finance* Author: Cole Prather Date: February 2026 Purpose: Develop the narrative arc, visual language, and production plan for illustrating/animating the "life" of a cash bubble in an economic fluid.

I. The Core Conceit

Your paper establishes the mathematics. This narrative gives it a *body*.

The idea: we follow a single **cash bubble** — a compressible gas bubble representing an investment position — from its birth (purchase) through its life (market fluctuations) to its death (liquidation). The bubble exists inside an **economic fluid** whose density encodes "value" — and from the relationship between bubble density and fluid density, an entire visual ecosystem emerges.

The Central Visual Metaphor

Imagine a tall column of fluid. The fluid is not uniform — it has a **density gradient** that increases with depth, like the ocean. This gradient *is* the economy:

Depth Region	Fluid Density	Economic Meaning
Surface (top)	Low density	High purchasing power / low inflation / "cheap" economy
Middle	Medium density	Moderate purchasing power / baseline conditions
Deep (bottom)	High density	Low purchasing power / high inflation / "expensive" economy

A bubble is injected into this column. Its density is fixed at the moment of creation (the **purchase price** determines its "composition"). Whether it floats, sinks, or hovers depends entirely on the relationship between its density and the surrounding fluid — which is to say, between its *entry value* and the *current economic conditions*.

This is not merely illustrative. It maps directly onto your existing mathematics:

$$d(t) = \ln \frac{P_{\text{entry}}}{P(t)}$$

- d > 0** → bubble density > fluid density → **sinks** (underwater position, loss)
- d = 0** → bubble density = fluid density → **neutral buoyancy** (break-even)
- d < 0** → bubble density < fluid density → **rises** (profit, positive buoyancy)

The density gradient of the fluid *is* the price history. The bubble's buoyancy *is* its P&L.

II. Narrative Arc — "The Life of a Cash Bubble"

The story unfolds in **seven acts**, each corresponding to a distinct physical and financial state. The narrative is designed to be modular — any act can stand alone as a demonstration, but together they tell a complete story.

Act 1: The Economic Fluid at Rest

Visual: A tall, transparent column of fluid. The fluid has a visible density gradient — perhaps rendered as a color gradient from light blue (top) to deep indigo (bottom). Small particles drift lazily, showing gentle convection currents (background market activity). A "depth gauge" on the side marks price levels.

Narration concept: *"Before any investment exists, there is the economy — a fluid medium through which all value flows. Its density is not uniform. Near the surface, dollars stretch far. In the depths, the same dollars buy less. This gradient is the cost of living, the weight of inflation, the pressure of prices."*

Physics: Establishes the fluid as a stratified medium with density $\rho(z)$, where z is depth. The density gradient $d\rho/dz$ encodes the inflationary environment. Horizontal flows represent cash circulation (your Q_{in} , Q_{out}). The "atmospheric pressure" at the surface is the baseline price level.

Data connection: The CREAM Data Analysis sheet provides the economic model context — income, spending, and the "fluid level" of cash accounts over 128 months. The BankTank simulation provides the multi-tank topology as the "plumbing" through which the fluid circulates.

Act 2: Birth — A Bubble Forms

Visual: A syringe-like injector (representing cash transfer from a liquid account) pushes a bolus of fluid

into the column. But the moment it enters the investment region, it *transforms* — the liquid becomes gas, forming a small, spherical bubble. The bubble's color/opacity encodes its entry price. It initially hovers at neutral buoyancy — its density matches the surrounding fluid at that exact depth.

Narration concept: *"A thousand dollars leaves the checking account — incompressible, certain, one dollar equals one dollar. But the moment it crosses the boundary into the market, something changes. The money becomes compressible. Its value is no longer fixed. It is now a bubble — buoyant, volatile, alive."*

Physics: This is the phase transition from incompressible liquid (cash) to compressible gas (investment). The bubble forms with:

- Volume $V_0 = n_0 \times P_0$ (initial dollar value)
- Internal "moles" n_0 = shares/coins purchased (conserved quantity)
- Pressure P_0 = entry price (initial equilibrium with surroundings)
- Temperature T_0 = current market sentiment

The ideal gas analogy: $PV = nRT$. At the moment of creation, the bubble is in thermal and mechanical equilibrium with its environment.

Data connection: Each Bitcoin purchase in the CREAM Bitcoin sheet (25+ transactions, \$1,145 total invested) represents a bubble birth event. Each stock purchase similarly. The "pre-fee" and "post-fee" columns show the actual cash-to-gas conversion, with fees as "heat of vaporization" — energy lost in the phase transition.

Act 3: Floating Life — The Market Breathes

Visual: Time passes. The bubble drifts. The surrounding fluid's density shifts — sometimes the gradient steepens (inflation), sometimes it relaxes (deflation/bull market). The bubble responds: it expands when surrounding pressure drops (price rises — the bubble's value grows), compresses when pressure increases (price falls — value shrinks). The bubble oscillates gently around its equilibrium depth, sometimes rising (profit), sometimes sinking (loss).

Multiple bubbles of different sizes and colors populate the column — a portfolio. Older bubbles are larger or smaller depending on market history. Newer bubbles are still small. Some cluster together (correlated assets), others drift independently.

Narration concept: *"The bubble does not choose its fate. The market breathes — expanding, contracting — and the bubble responds. When prices rise, the surrounding pressure drops, and the bubble expands. Its value grows. It rises. When prices fall, pressure increases, the bubble compresses, and it sinks toward the depths. The shares inside never change. Only the bubble's size — its dollar value — fluctuates with every market breath."*

Physics: This is the core thermodynamic life of the bubble. The key equation is:

$$\Delta U = \delta Q - \delta W$$

where:

- ΔU = change in bubble's internal energy (unrealized gain/loss)
- $\delta Q = n \, dP$ = "heat" absorbed from market appreciation
- $\delta W = -P \, dn$ = "work" extracted via selling

During a holding period (no transactions), $dn = 0$, so $\delta W = 0$ and all energy change is "heat exchange" with the market — pure unrealized gains/losses. The bubble is a passive thermometer of market conditions.

Sub-bubble dynamics: Following your Section 6.4 extension, each purchase lot is a *sub-bubble* within the larger investment bubble. The portfolio is a *distribution of depths*. A Bitcoin portfolio with 25 purchase lots has 25 sub-bubbles, each at a different depth relative to the current price surface.

Data connection: The CREAM Bitcoin sheet tracks this exactly — each row is a sub-bubble with its own entry price, creating the depth distribution $d(t) = \ln(P_{\text{entry}} / P(t))$. The "Current Market Value" field (\$69,774.71 at last snapshot) sets the "fluid level" that determines which sub-bubbles are above water and which are below.

Act 4: Expansion — The Bull Run

Visual: The fluid density around the bubble drops rapidly (prices surge). The bubble expands dramatically — growing larger, becoming more buoyant, rising toward the surface. Its color shifts from neutral to warm/golden. The bubble reaches the "surface" — maximum buoyancy, maximum unrealized profit.

At the surface, the bubble is enormous compared to its birth size. It pushes against the surface tension (psychological resistance to selling at the peak). The surface ripples and deforms.

Narration concept: *"In a bull market, the surrounding pressure collapses. The bubble expands — freely, exuberantly. Its volume doubles, triples. The same shares now represent enormous value. The bubble rises toward the surface, straining against the boundary between 'paper profit' and 'real money.' The temptation to extract is immense. But the First Law is impartial: work can only be extracted by letting gas escape — by selling."*

Physics: This is isothermal expansion at high temperature. If market sentiment T stays elevated:

$$PV = nRT \implies V = \frac{nRT}{P}$$

As P decreases (relative to entry), V increases proportionally. The bubble stores potential energy — unrealized gains. The surface tension analogy (transaction costs, psychological barriers, tax implications) resists "popping."

Data connection: The CREAM data captures a real bull run — Bitcoin rising from ~\$9K (May 2020) through \$60K+ peaks. The bubble expansion is empirically traceable through the monthly market value column.

Act 5: Work Extraction — Selling as Controlled Deflation

Visual: A valve opens at the top of the bubble. Gas escapes — but as it crosses back through the surface into the liquid region, it transforms back into incompressible fluid. Cash. The escaping gas does *work* on its surroundings: it pushes fluid into the liquid accounts (checking, savings). The bubble shrinks as gas is released, but it doesn't vanish — some gas remains.

Narration concept: *"Selling is not destruction. It is work extraction. The investor opens a valve and lets the compressed gas escape — but as it crosses the phase boundary back into the liquid domain, it becomes cash again. Incompressible. Certain. The proceeds flow into the checking account, raising the liquid level. The bubble shrinks, but every dollar extracted is a dollar of real, crystallized work."*

Physics: This is the core thermodynamic work equation:

$$W = P \cdot \Delta n = P \cdot (n_1 - n_2)$$

Isobaric (constant-price) selling: work = price $\times$ shares sold. The cash extracted is literal thermodynamic work — organized energy transferred from the compressible system to the incompressible one.

For partial selling, the bubble shrinks but persists. For full liquidation, the bubble collapses entirely — all gas converts back to liquid.

Data connection: The CREAM Bitcoin "Sold" transactions show explicit work extraction events. The "Liquid Profit" field (\$204.55 at snapshot) represents the cumulative thermodynamic work extracted from the Bitcoin bubble over its lifetime. The BankTank "Reservoir $\rightarrow$ Pool" flow channels model where the extracted cash goes.

Act 6: Compression — The Bear Market / Crash

Visual: The fluid density surges (prices crash). The bubble is squeezed violently — compressing, sinking, its color shifting to cold blue. Sub-bubbles that were near the surface plunge deep. Some sub-bubbles reach the bottom of the column (total loss threshold). The bubble oscillates wildly — jerk is visible as rapid changes in the direction of compression.

Two contrasting scenarios play out side by side:

1. **Adiabatic crash:** Rapid compression, no time for heat exchange. $PV^\gamma = \text{constant}$. The bubble heats internally (panic) but cannot equilibrate. Forced selling at the bottom extracts minimal work.
2. **Isothermal weathering:** Slow compression, the bubble equilibrates with its surroundings. Patient holding. The bubble compresses but maintains thermal contact — it can re-expand when conditions change.

Narration concept: *"A crash is compression — sudden, violent, indiscriminate. The fluid thickens. Pressure mounts from all sides. The bubble shrinks. Sub-bubbles that once floated freely now sink past neutral buoyancy, past break-even, into the red depths. The physics is clear: those who sell during adiabatic compression — panic sellers — extract the least work. The gas escapes at maximum pressure and minimum volume. Those who endure the isothermal path — who let the bubble equilibrate — preserve the option to expand again."*

Physics: Adiabatic process: $PV^\gamma = \text{const} \rightarrow$ rapid compression with no market equilibration. This models panic selling during flash crashes. Isothermal process: $PV = nRT$ at constant $T \rightarrow$ slower compression, investor maintains "thermal contact" (stays informed, doesn't panic).

The Carnot efficiency bound emerges naturally: the maximum fraction of market appreciation capturable as realized profit is bounded by:

$$\eta_{\text{Carnot}} = 1 - \frac{T_{\text{cold}}}{T_{\text{hot}}} = 1 - \frac{P_{\text{buy}}}{P_{\text{sell}}}$$

Data connection: The 2022 crypto winter is visible in the CREAM data — Bitcoin dropping from ~\$47K to ~\$16K. The monthly market value column shows the bubble compressing in real time over ~12 months.

Act 7: Death, Rebirth, or Equilibrium

Three possible endings, each a distinct thermodynamic fate:

7a — Full Liquidation (Bubble Death): The entire bubble is drained. All gas converts to liquid. The bubble ceases to exist. The liquid level in the cash tanks rises by the extracted work. A gravestone marker shows the entry price, exit price, and efficiency (actual return / Carnot bound).

7b — Eternal Hold (Frozen Bubble): The bubble is never sold. It drifts indefinitely, expanding and contracting with market cycles. Its value oscillates but no work is ever extracted. In the limit, it becomes part of the fluid's background — an estate asset, a frozen thermodynamic state.

7c — DCA Rebirth (New Sub-Bubbles): The investor injects new cash periodically. New sub-bubbles form at current market depth. The portfolio bubble becomes a composite — a *distribution* of sub-bubbles at various depths. Dollar-cost averaging creates a uniform depth distribution over time, naturally hedging against any single entry point.

Narration concept: *"Every bubble eventually faces one of three fates. Full extraction — the bubble dies, but its work lives on as liquid cash. Eternal suspension — the bubble drifts forever, a store of unrealized potential, never converted to certainty. Or continuous rebirth — new sub-bubbles injected regularly, spreading the depth distribution, averaging the risk. The physics favors no ending. Only the investor chooses."*

III. The Ecosystem — Beyond a Single Bubble

Once the single-bubble narrative is established, the framework naturally extends to describe the *ecosystem* of an entire financial life:

Inflation as Fluid Densification

Inflation increases the fluid's density everywhere. All bubbles that were neutrally buoyant now find themselves relatively lighter — their *nominal* value may be unchanged, but their *real* purchasing power has declined. Cash at the bottom of a tank (uninvested savings) is most affected — it doesn't compress or expand, it just... sits there, while the world around it gets heavier.

Visual: The color gradient deepens over time. Bubbles that were once at equilibrium slowly begin to sink — not because they changed, but because the fluid changed around them.

Interest Rates as Pressure Gradients

Higher interest rates increase the "pressure" in the fluid — compressing all bubbles (lowering asset valuations). Lower rates reduce pressure — allowing bubbles to expand (higher valuations). The Fed is a pressure regulator.

Different Asset Classes as Different Gases

Not all bubbles are the same gas. Bitcoin bubbles have a high γ (adiabatic index) — they compress and expand violently. Bond bubbles have a low γ — they barely respond to pressure changes. Stock bubbles are somewhere in between. The gas composition determines the bubble's response function to market pressure changes.

The Multi-Tank Topology as Plumbing

The BankTank model provides the "plumbing" through which liquid flows between tanks (accounts). The bubbles float within specific tanks (investment accounts), but the liquid circulates through the entire system. Income flows in from external sources (a faucet). Spending drains out (a valve at the bottom). The system's health is monitored by flow velocity, acceleration, and jerk — the signal dynamics from your paper.

IV. Visualization Production Plan

The narrative above demands different levels of visual fidelity for different purposes. Here is a tiered production plan, from simplest to most ambitious:

Tier 1: Static Figures for the Paper (Excel/Python → PDF)

Purpose: Publication-quality diagrams for the academic paper. **Tools:** Python (matplotlib/plotly) or Excel

charts, exported as high-resolution images. **Deliverables:**

Figure	Content	Data Source
Fig. 1 — Tank Schematic	Multi-account topology with flow arrows	BankTank_v5 structure
Fig. 2 — Bubble Diagram	Single bubble in stratified fluid with depth/density annotations	Conceptual (hand-designed)
Fig. 3 — P-V Diagram	Pressure-volume cycle for a Bitcoin position (buy-hold-sell)	CREAM Bitcoin sheet
Fig. 4 — Sub-Bubble Depth Distribution	Scatter/histogram of purchase lots by depth d(t)	CREAM Bitcoin transaction data
Fig. 5 — Signal Dynamics Time Series	Velocity, acceleration, jerk of net worth over 128 months	CREAM Data Analysis sheet
Fig. 6 — Phase Portrait	V vs. dV/dt with four-quadrant annotation	CREAM Data Analysis sheet

Estimated effort: 2–3 sessions. Mostly data extraction and formatting.

Tier 2: Animated Explanatory Videos (Manim)

Purpose: Conference talks, YouTube companion, classroom demonstrations. **Tools:** Manim (3Blue1Brown's animation library, Python-based). **Why Manim:** It excels at mathematical animation — equations that transform, objects that move according to physical laws, camera pans through coordinate systems. It is the natural choice for a physics-education audience. Grant Sanderson's aesthetic signals "rigorous but accessible."

Deliverables:

Animation	Duration	Content
Anim 1 — "The Fluid"	60–90 sec	Establishing shot: density gradient, particle flow, depth gauge. Introduce the economic fluid concept.
Anim 2 — "Birth of a Bubble"	90–120 sec	Cash injection → phase transition → bubble formation. Show $PV = nRT$ equation appearing alongside.
Anim 3 — "Market Breathing"	120–180 sec	Bubble expanding/contracting with price oscillations. Split screen: bubble animation + real BTC price chart.
Anim 4 — "Work Extraction"	90–120 sec	Selling as valve opening. Gas → liquid conversion. Cash level rising in tank. $W = P\Delta n$ equation.
Anim 5 — "The Crash"	120–180 sec	Adiabatic vs. isothermal compression side-by-side. Carnot efficiency bound.
Anim 6 — "The Portfolio"	120–180 sec	Multiple sub-bubbles at different depths. DCA as regular injection. Depth distribution histogram building in real time.
Anim 7 — "Full Lifecycle"	300–600 sec	Complete narrative, Acts 1–7, with real CREAM data driving the bubble's behavior.

Estimated effort: 3–6 weeks for full suite. Anim 1–2 achievable in a single focused weekend with Manim experience.

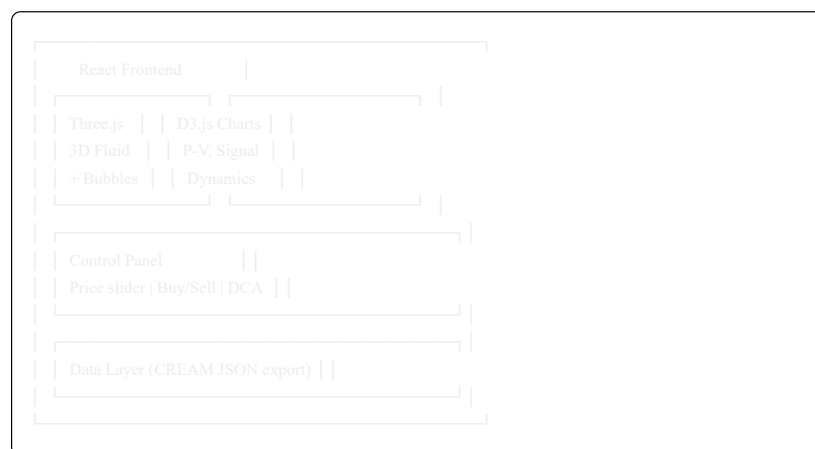
Tier 3: Interactive Web Application (React/Three.js or D3.js)

Purpose: Website companion, educational tool, live demonstration. **Tools:** React frontend + Three.js (3D fluid/bubble rendering) or D3.js (2D data-driven visualization). Could also use p5.js for a more "sketch-like" interactive feel.

Features:

Feature	Description	Priority
Fluid column	Animated stratified fluid with adjustable density gradient	Core
Bubble injection	Click to "buy" — creates a bubble at current depth	Core
Market price slider	Adjust "pressure" — watch bubbles expand/compress in real time	Core
Sell valve	Click to extract work — bubble shrinks, cash level rises	Core
P-V diagram	Real-time P-V trace as user interacts with bubble	Core
Sub-bubble view	Toggle to see individual purchase lots within a portfolio bubble	Extended
Time playback	Load CREAM data and replay 128 months of real financial history	Extended
Signal dashboard	Live velocity, acceleration, jerk gauges	Extended
Inflation control	Adjust fluid density gradient over time	Extended
Scenario comparison	Side-by-side: DCA vs. lump sum, panic sell vs. hold	Aspirational

Architecture:



Estimated effort: 4–8 weeks for core features. Suitable as a semester-long student project or personal build.

Tier 4: Classroom Physical Demonstration

Purpose: Live lecture demo. A real fluid column with real bubbles. **Setup:** A tall graduated cylinder filled with a density-stratified fluid (saltwater gradient or sugar solution). Inject bubbles of different densities (air, CO₂, or dyed oil droplets) and watch them find their equilibrium depth.

Connection to theory: "This bubble was injected at a density of 1.02 g/cm³. The fluid at this depth has a density of 1.03 g/cm³. The bubble sinks. If I dilute the fluid — analogous to a price increase — watch: the bubble rises. Its 'value' increased without anything changing about the bubble itself."

Estimated effort: One afternoon of setup. Deeply memorable for students.

V. Production Sequencing — What to Build When

Given the project's current state (~70% toward publication-ready), the recommended build order prioritizes what serves the paper first, then extends outward:

Phase 1: Paper Figures (Immediate — next 1–2 weeks)

Build Tier 1 static figures. These are needed regardless of all other decisions. The P-V diagram (Fig. 3) and sub-bubble depth distribution (Fig. 4) are the most novel and the most directly connected to this narrative framework. These also serve as "proof of concept" that the data supports the visual metaphor.

Phase 2: Single Manim Animation (Short-term — next 2–4 weeks)

Build Anim 2 ("Birth of a Bubble") as a standalone proof-of-concept. This is the most visually striking single scene and the one most likely to generate interest if shared. It exercises the core visual language (fluid, bubble, phase transition) without requiring the full narrative arc.

Phase 3: Interactive Prototype (Medium-term — 1–2 months)

Build the Tier 3 React app with core features only (fluid column, bubble injection, price slider, sell valve).

This becomes the "playground" for the concept — shareable, interactive, and immediately engaging for anyone who encounters the work.

Phase 4: Full Animation Suite (Long-term — 2–4 months)

Complete the Manim animation series (Anims 1–7). This is the full "film" version of the narrative and serves as the definitive explanatory companion to the paper.

Phase 5: Classroom Demo (Whenever convenient)

Build the physical demonstration. This can happen at any time and is independent of the digital production pipeline.

VI. Technical Notes & Mapping Reference

Density-Buoyancy-Profit Mapping (Complete)

Physical Quantity	Symbol	Financial Quantity	Units
Fluid density at depth z	$\rho_{\text{fluid}}(z)$	Current price level / CPI	\$/unit
Bubble density	ρ_{bubble}	Entry price (cost basis)	\$/share
Buoyancy force	$F_b = (\rho_{\text{fluid}} - \rho_{\text{bubble}})gV$	Unrealized P&L	\$
Depth coordinate	$d = \ln(P_{\text{entry}}/P_{\text{now}})$	Log return (negative = profit)	dimensionless
Neutral buoyancy depth	$d = 0$	Break-even point	—
Surface tension	σ	Transaction costs + tax friction	\$/trade
Viscosity	μ	Market liquidity (bid-ask spread)	\$/time
Temperature	T	Market sentiment / volatility	(P ² /R)
Pressure	P	Price per share	\$/share
Volume	$V = nP$	Dollar value of position	\$
Moles (conserved)	n	Shares/coins held	shares
Heat	$\delta Q = n \, dP$	Unrealized gain from price change	\$
Work	$\delta W = -P \, dn$	Cash extracted via selling	\$
Adiabatic index	γ	Asset class volatility response	dimensionless

Key Equations for Animation Logic

Bubble position (depth):

$$z(t) = z_0 + \int_0^t v_b(\tau) \, d\tau$$

where v_b is the bubble's rise/fall velocity, governed by the balance of buoyancy and drag:

$$m_b \frac{dv_b}{dt} = (\rho_{\text{fluid}}(z) - \rho_{\text{bubble}})gV_b - \frac{1}{2}C_D \rho_{\text{fluid}} A v_b^2$$

For animation purposes, overdamped motion (terminal velocity reached quickly) simplifies to:

$$v_b \propto (\rho_{\text{fluid}}(z) - \rho_{\text{bubble}}) \cdot V_b$$

Positive when fluid is denser (bubble rises = profit). Negative when bubble is denser (sinks = loss).

Bubble radius (visual size):

$$R(t) = \left(\frac{3nR_{\text{gas}}T}{4\pi P(t)} \right)^{1/3}$$

Radius scales with P^(-1/3) — as price drops, bubble compresses. As price rises, bubble expands.

VII. Open Questions for Development

- 1. **Color encoding:** Should bubble color represent entry price (fixed at birth) or current P&L status (dynamic)? The latter is more informative but less physically motivated.
- 2. **2D vs. 3D:** Should the fluid column be a 2D cross-section (simpler, more diagram-like, better for Manim) or a 3D rendered volume (more immersive, better for Three.js webapp)?
- 3. **Time scale:** Should animations run in "real time" mapped to monthly data, or should dramatic events (crashes, bull runs) be compressed/expanded for narrative effect?
- 4. **Sound design:** For the Manim animations, should the bubble's expansion/compression have an associated pitch (frequency $\propto$ size, like a resonant cavity)? This would make market volatility audible.
- 5. **Multiplayer/classroom mode:** Should the Tier 3 webapp support multiple users injecting bubbles simultaneously (each student tracks their own "portfolio" in real time against the same market)?

VIII. Connection Back to the Paper

This narrative framework is not a departure from the academic work — it is its *pedagogical arm*. Every visual element maps to a specific equation in the paper. Every animation act corresponds to a section of the theoretical development. The progression mirrors the paper's structure:

Narrative Act	Paper Section	Core Equation
Act 1: Economic Fluid	§II: Fluid Finance Framework	$dV/dt = Q_{in} - Q_{out}$
Act 2: Bubble Birth	§III.1: Compressibility of Investment Value	$V = nP$
Act 3: Floating Life	§III.2–3: Ideal Gas Law + First Law	$PV = nRT; \Delta U = \delta Q - \delta W$
Act 4: Expansion	§III.3: Realized Gains as Work	$W = P\Delta n$
Act 5: Work Extraction	§III.4: Adiabatic vs. Isothermal	$PV^\gamma = \text{const vs. } PV = nRT$
Act 6: Crash	§VI.3: Volatility-Price Scaling	$P^2 = RT$
Act 7: Endings	§VI.4: Sub-bubble Depth Dynamics	$d(t) = \ln(P_{entry}/P(t))$
Ecosystem	§IV: Signal Dynamics + Control Theory	$v, a, j; \text{PID interpretation}$

The goal is that someone who watches the animation and then reads the paper feels *recognition* — "Oh, that's what that bubble was doing" — and someone who reads the paper and then watches the animation feels *understanding* — "Oh, that's what the equations mean."

This framework is a living document. It will evolve as production begins and as the paper approaches its final form.